

FINITE CONTEXT IS NOT PERSISTENT STATE: A FROZEN FALSIFICATION STUDY IN A LEWORLDMODEL-DERIVED JEPA

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ABSTRACT

Pixel joint-embedding predictors can be temporally causal and still forget: evidence disappears the moment it leaves a finite context window. We test whether an explicit persistent state repairs this in a LeWorldModel (LeWM)-derived pixel JEPA. The candidate, a compact Shared-Action Shrinkage Predict–Correct (SAS-PC) module, adds two action-conditioned predict–correct states to a true sliding $H = 3$ predictor; it receives no simulator state, reward, corruption label, or memory-specific objective. Before opening the cohort we froze five DeepMind Control tasks, eight separately trained designs, five seeds, 100 epochs, four held-out corruption families, and eleven conjunctive gates. The 200-cell confirmation **fails**. SAS-PC changes held-out prior-state error (positive favors SAS-PC) by **+12.44%** versus no carrier, **-2.10%** versus the per-cell better GRU/SSM (95% crossed-bootstrap CI [-13.35%, +9.11%]), and **-38.12%** versus a causally legal initial-frame/action integrator. Its action-transport intervention passes at **+9.48%**, while the joint-read contrast is **-5.39%**. Swimmer and Acrobot take opposite signs in all five seeds against the recurrent envelope, every integrator comparison is unfavorable, and representation-rank and convergence guards fail. The favorable no-carrier and action-transport contrasts therefore cannot be promoted to confirmed class or mechanism claims. Finite context is not persistent state, but that architectural fact is not itself a performance result.

1 INTRODUCTION

Joint-Embedding Predictive Architectures (JEPAs) learn dynamics by forecasting representations rather than reconstructing pixels. Recent systems add action conditioning, short video context, and latent-space planning (Maes et al., 2026; Ghaemi et al., 2025; Zhou et al., 2025; Assran et al., 2025), but their temporal states are not interchangeable, and three regimes are worth separating. A frame-local encoder carries no temporal state at all. A causal predictor over the latest H latent/action tokens has *finite-context* memory: it can integrate evidence inside the window, but whatever leaves the window is unrecoverable. An episode state $m_t = F(m_{t-1}, z_t, a_{t-1})$ can, in principle, *persist* evidence indefinitely. None of these properties implies causal representation learning (Schölkopf et al., 2021).

Published LeWM is action-conditioned and causally masked, with configured history $H = 3$ for PushT and OGBench-Cube and $H = 1$ for TwoRoom (Maes et al., 2026); it is therefore neither memoryless nor non-causal. The narrower question this paper asks is whether an explicit state surviving beyond H transports *useful* information under partial observability. Because any recurrent module beats a windowed baseline whenever old evidence has value, a candidate carrier must also survive ordinary recurrent baselines, a causally legal initial-frame/action summary, exact component interventions, and representation and convergence health checks.

We integrate compact Shared-Action Shrinkage Predict–Correct (SAS-PC) memory into a causally normalized, VICReg-style (Bardes et al., 2022) LeWM-derived pixel host. All eight designs share the encoder, the $H = 3$ predictor, data, targets, optimization, and evaluation; only the persistent carrier or a named intervention changes. Because the objective replaces LeWM’s SIGReg recipe (Balestriero & LeCun, 2025), this is an architecture study, **not** evidence that SAS-PC improves original LeWM.

The result is deliberately sharper than “memory helps.” Favorable finite-window and action-transport contrasts coexist with unfavorable stronger references, unsupported hierarchy contrasts, and failed validity guards; task, seed, corruption, and phase decompositions show why these statements must remain separate. Our contributions are: (i) an explicit within-grid integration separating finite context from persistent state, with the carrier’s update equations and design rationale in full; (ii) a frozen 5-task $\times$ 8-design $\times$ 5-seed confirmation with recurrent, integrator, intervention, health, and convergence controls; and (iii) a complete, task- and seed-resolved falsification identifying exactly which registered contrasts pass and which fail.

2 RELATED WORK AND CLAIM SCOPE

Finite-context JEPAs include DINO-WM, seq-JEPA, and LeWM (Maes et al., 2026; Ghaemi et al., 2025; Zhou et al., 2025); a Transformer over a configured block need not preserve evidence after the block is discarded. Persistent state is well established in PlaNet/Dreamer recurrent state-space models (RSSMs), recurrent and S4 world-model comparisons, and long-horizon memory studies (Hafner et al., 2019; 2020; 2025; Deng et al., 2023; Samsami et al., 2024). Multi-timescale state and predict–correct filtering predate SAS-PC in MTS3 and recurrent Kalman networks, and ELMUR and Flow Equivariant World Models provide other structured memories (Shaj et al., 2023; Becker et al., 2019; Shaj et al., 2021; Cherepanov et al., 2026; Lillemark et al., 2026). We therefore claim neither the first recurrent world model nor any novelty for recurrence, filtering, action conditioning, or multiple timescales in isolation.

Our claims form a strict ladder. The information-flow graph (Figure 1) establishes an **architectural** distinction between finite context and persistent state. Absence of future inputs establishes **temporal causality**, not causal representation. A registered performance contrast tests **state utility**; a separately retrained component control tests an implemented **mechanism**. None identifies environment causal structure; we evaluate the first four levels and use “causal” only for temporal information flow or an explicit model intervention. The full claim boundary is registered in Appendix G.

3 ARCHITECTURE: A PERSISTENT CARRIER INSIDE A FINITE-CONTEXT HOST

Every design uses a per-frame Vision Transformer (ViT) encoder E_θ (64×64 RGB, 8×8 patches, $D = 128$, six layers, four heads), an action-conditioned causal Transformer predictor P_ϕ (four layers, eight heads), and the true sliding window of the latest $H = 3$ latent/action tokens (Figure 1). Two synchronized views share encoder weights: corrupted frames produce the recurrent input $z_t^c = E_\theta(o_t^c)$, while clean frames produce active, non-stop-gradient targets $z_t^* = E_\theta(o_t)$ with dropout disabled. Per-frame feature normalization prevents future-frame or cross-example leakage; target statistics couple samples only through the loss. For valid aligned windows $\mathcal{W}$, the host objective is

$$\mathcal{L} = \mathbb{E}_{(i,t) \in \mathcal{W}} \left[\frac{1}{D} \|P_\phi(\tilde{z}_{i,t-2:t}^c, a_{i,t-2:t}) - z_{i,t+1}^*\|_2^2 \right] + \mathcal{L}_{\text{var}}(Z^*) + \mathcal{L}_{\text{cov}}(Z^*), \quad (1)$$

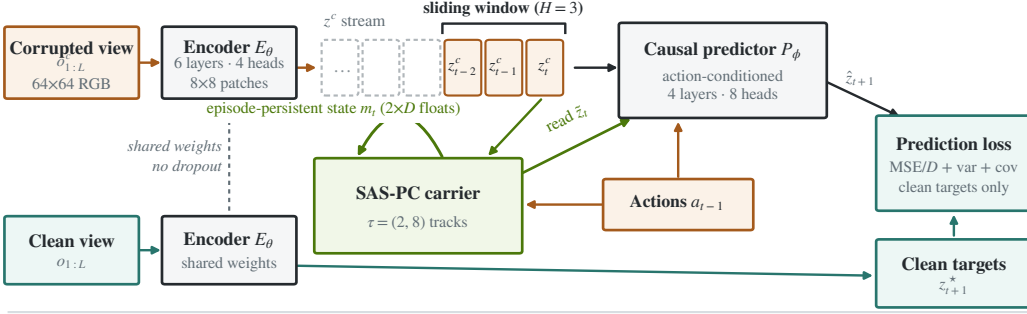
with unit-weight VICReg-style variance and covariance terms applied *only to active clean targets* (exact forms in Appendix A). Regularizing targets rather than predictions stabilizes the joint-embedding fixed point without giving the memory path an auxiliary teacher: there is no memory-specific loss, no hidden-clean update, and no state, reward, or corruption-label input anywhere in the model. This VICReg-style objective differs from original SIGReg LeWM (Maes et al., 2026; Balestriero & LeCun, 2025; Bardes et al., 2022); Section 2 scopes every claim accordingly.

3.1 THE CARRIER: PREDICT, CORRECT, READ

SAS-PC maintains two full-width states $m_t^k \in \mathbb{R}^D$, $k \in \{f, m\}$, with fixed structural timescales $\tau = (2, 8)$ and rates $\beta_k = 1 - e^{-1/\tau_k}$. Each step factors into three stages.

Predict: shared action transport. A single bias-free map W_a splits the last action into a multiplicative channel gate and an additive drive, $(d_{t-1}, v_{t-1}) = W_a a_{t-1}$, and extrapolates both states:

(a) Pixel JEPA host with an episode-persistent carrier



(b) One SAS-PC step: predict → correct → read

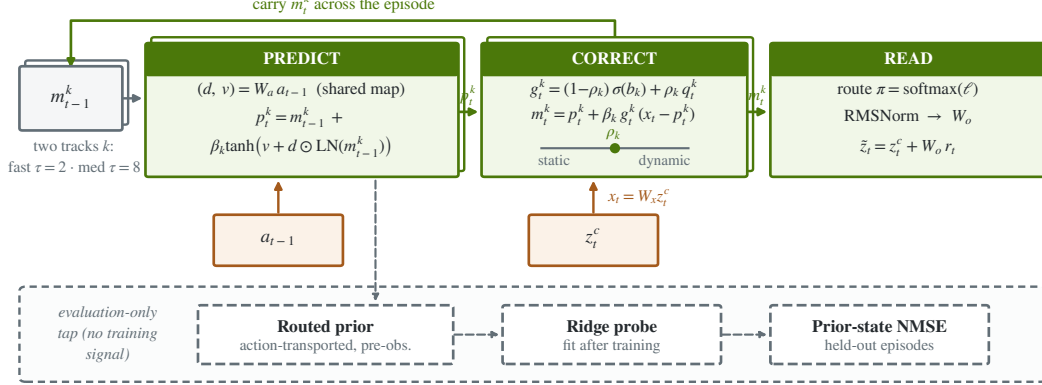


Figure 1: SAS-PC adds an episode-persistent path while the host predictor retains a finite $H = 3$ window. Corrupted and clean streams share the encoder, but only corrupted latents update memory. Green nodes are the SAS-PC predict–correct path; teal denotes clean targets and losses. Solid edges are model or training flow; the dashed strip at the bottom is the pre-correction tap that exists only for frozen post-training evaluation.

$$p_t^k = m_{t-1}^k + \beta_k \tanh(v_{t-1} + d_{t-1} \odot \text{LN}(m_{t-1}^k)). \quad (2)$$

Equation 2 advances each state as a bounded-drive integrator: β_k is the discrete-time update rate of a process with time constant τ_k , so the fast state ($\tau_f = 2$) tracks within-window transients while the medium state ($\tau_m = 8$) spans the 6–12-step training gaps and reaches toward the held-out 16–24-step freezes. The rates are fixed scalars by design: with learned rates, the rate, gate, and shrinkage parameters become mutually non-identifiable, and learned per-channel rate spectra underperformed two fixed scalars during development on disjoint tasks. The $\tanh$ bounds the per-step action displacement so blind rollouts cannot diverge, and $d_{t-1} \odot \text{LN}(m_{t-1}^k)$ makes transport state-dependent rather than a constant drift; both levels share one physical W_a because level-specific heads lost to a shared-action control in the same disjoint-task development. During a corruption gap this term supplies the only forward dynamics acting on the belief (correction still fires, but on a corrupted latent); it is exactly the component the no-action arm deletes, elevating action transport from a design choice to a registered, testable mechanism.

Correct: shrinkage-gated innovation. The current corrupted latent is projected once, $x_t = W_x z_t^c$, and each prior is corrected by its innovation $x_t - p_t^k$:

$$\begin{aligned} q_t^k &= \sigma(b_k + [w_z^\top \text{LN}(z_t^c) + w_e^\top \text{LN}(x_t - p_t^k)] / \sqrt{D}), \\ g_t^k &= (1 - \rho_k) \sigma(b_k) + \rho_k q_t^k, \quad m_t^k = p_t^k + \beta_k g_t^k (x_t - p_t^k), \quad \rho_k = \sigma(c_k). \end{aligned} \quad (3)$$

The gate g_t^k plays the role of a Kalman gain (Becker et al., 2019), but with two deliberate restrictions. First, the correction is *rate-scaled*: multiplying by β_k caps each level’s observation intake at its structural timescale, a shrinkage that prevents a slow state from being overwritten by one noisy frame. Second, g_t^k is a convex mixture of a constant “static expert” $\sigma(b_k)$, which trusts observations by a fixed amount, and an input-conditioned “dynamic expert” q_t^k , which reads the latent and the innovation.

The learned scalar ρ_k selects a point on this segment; $\rho_k \rightarrow 0$ recovers purely static correction and $\rho_k \rightarrow 1$ purely dynamic correction, which are exactly the frozen static and dynamic endpoint arms. Development on disjoint tasks showed static winning on some tasks and dynamic on others, so SAS-PC learns the interpolation instead of committing globally; whether the learned point beats the per-cell better endpoint is the shrinkage gate of Table 2. The mixture also encodes a falsifiable prediction about corruption type: when observations are merely *absent* (freezes), correction is cheap and transport dominates, but when the incoming latent is itself *corrupted* (noise, checkerboard), an input-conditioned gate must learn to close; Section 5’s corruption decomposition tests this.

Read: routed residual with a null initialization. A single time-independent softmax combines the corrected states, and a residual returns the result to the predictor stream:

$$\tilde{z}_t^c = z_t^c + W_o \text{RMSNorm}(\pi_f m_t^f + \pi_m m_t^m), \quad \pi = \text{softmax}(\ell). \quad (4)$$

The route is deliberately not a per-token router: any dynamic-memory claim must then live in action evolution or correction timing, where the interventions can test it, rather than in a confounded learned horizon selector. RMSNorm anchors the read scale so state magnitude cannot act as a shortcut, and the residual form leaves the predictor interface unchanged, making the no-carrier host an exact ablation. Initialization is identity/zero ($W_x = I$, $W_o = W_a = 0$, uniform route, $\rho_k = \frac{1}{2}$): at step zero the fused model *is* the no-carrier host, so every memory effect is learned and all paired arms start from the same prediction path; both states warm-start from $W_x z_0^c$. The module carries $2D$ persistent floats and $2D^2 + 2AD + 2D + 6$ trainable parameters — 33,286–36,614 across task action dimensions, versus 35,048 for the width-matched GRU; recurrent carriers differ by at most 5.29% and total models by at most 0.09%.

3.2 MATCHED CARRIERS AND EXACT INTERVENTIONS

Because every stage of Equations 2–4 is a separable design decision, each frozen arm deletes or pins exactly one of them (Table 1). The GRU (gated recurrent unit) and diagonal state-space model (SSM) carriers read visual latents only, while the shared predictor remains action-conditioned for all arms; the no-action arm therefore isolates *internal* action transport within the candidate family rather than action information in general. All arms are separately trained from scratch; none is a post hoc modification of a shared checkpoint.

4 FROZEN EVALUATION

4.1 COHORT, CORRUPTIONS, AND GRID

The unopened cohort contains Acrobot Swingup, Manipulator Bring Ball, Quadruped Run, Stacker Stack-4, and Swimmer-15 from DeepMind Control (Tassa et al., 2018) (hereafter Acrobot, Manipulator, Quadruped, Stacker, Swimmer). Each task supplies 1,200 training and 240 validation episodes of length 48 with disjoint random-action trajectories; native task state is never a training input. Training corrupts a 6–12-step interval by mean replacement or spatial cutout. Evaluation uses four *unseen* corruption families: frozen frame (freeze), Gaussian noise, checkerboard replacement, and 16–24-step long freezes. We froze 5 tasks $\times$ 8 designs $\times$ 5 optimizer seeds = 200 cells, each trained with AdamW (learning rate 3×10^{-4} , weight decay 10^{-5} , batch 64) for exactly 100 epochs —

Table 1: The eight frozen arms as equation-level interventions. Every arm retrains the full model; only the named component changes. Parameter receipts are in Appendix A.

Arm	Change relative to Eqs. (2)–(4)	Question it isolates
No carrier	carrier deleted ($\tilde{z}_t^c = z_t^c$)	is any episode state useful?
GRU	carrier $\rightarrow$ parameter-matched GRU on z_t^c	does generic recurrence suffice?
Diag. SSM	carrier $\rightarrow$ learned diagonal SSM on z_t^c	does linear recurrence suffice?
SAS-PC	none (full candidate)	the candidate
No action	$W_a \equiv 0$ in Eq. (2)	internal action transport
Single read	$\pi = (0, 1)$ pinned in Eq. (4)	value of the two-state read
Static	$\rho_f = \rho_m = 0$ in Eq. (3)	input-independent correction
Dynamic	$\rho_f = \rho_m = 1$ in Eq. (3)	fully input-conditioned correction

Table 2: Frozen confirmation gates and observed receipts. Decisions use unrounded values.

Gate	Registered requirement	Observed	Verdict
Integrity	200/200 valid cells	200/200	PASS
Recurrent envelope	$\geq 3\%$; $CI > 0$; $\geq 18/25$; $\geq 4/5$	-2.10%; [-13.35%, +9.11%]; 11/25; 2/5	FAIL
No carrier	$\geq 5\%$; $\geq 20/25$; $\geq 4/5$	+12.44%; 24/25; 5/5	PASS
Integrator	$\geq 3\%$; $\geq 18/25$; $\geq 4/5$	-38.12%; 0/25; 0/5	FAIL
Deep gap	$CI > 0$; $\geq 3/5$	-1.74%; [-12.95%, +9.46%]; 2/5	FAIL
Action transport	$\geq 5\%$; $CI > 0$; $\geq 18/25$; $\geq 4/5$	+9.48%; [+2.88%, +16.09%]; 23/25; 5/5	PASS
Joint read	$\geq 3\%$; $CI > 0$; $\geq 18/25$; $\geq 4/5$	-5.39%; [-13.12%, +0.67%]; 6/25; 1/5	FAIL
Shrinkage	mean and $CI \geq -1\%$	-1.89%; [-4.88%, +0.37%]	FAIL
Clean guard	degradation $\leq 3\%$	effect +11.73%	PASS
Representation	$\text{var} \geq 10^{-4}$; rank ≥ 16 ; 200/200	var 0.0228 (200/200); rank 2.02 (144/200)	FAIL
Convergence	$ \text{late change} \leq 5\%$; 200/200	max +132.09%; 126/200	FAIL

no early stopping, best-checkpoint selection, task or seed exclusion, rescue sweep, or architecture revision; exact optimization, parameter, and hash receipts are in Appendix A.

4.2 PRIOR-STATE ENDPOINT AND CONSERVATIVE REFERENCES

After training, ridge probes ($\lambda = 10^{-3}$) map each model coordinate to the native task observation; labels appear only in this frozen evaluation layer. The primary coordinate is the **prior before the current observation** — the $H = 3$ predictor prior for no carrier, the previous hidden read for GRU, the transition prior for SSM, and the action-transported routed prior for SAS-PC — because a pre-correction prior is the only coordinate that measures information *transported through* missing observations rather than read off the current frame. Per-coordinate normalized mean-squared error (NMSE) standardizes each target dimension by its clean-training standard deviation, and the headline averages deep-gap plus first-post samples within each held-out corruption, then averages the four corruptions. For candidate error c_{ts} and reference error r_{ts} on task t and seed s ,

$$\delta_{ts} = \frac{r_{ts} - c_{ts}}{\max(|r_{ts}|, 10^{-12})}, \quad (5)$$

with **positive favoring SAS-PC** and tasks and seeds weighted equally. Two references are deliberately conservative. The *recurrent envelope* selects the lower of separately trained GRU/SSM per task–seed cell on the primary metric and reuses that fixed identity for deep and clean checks, so the candidate faces the stronger baseline in every cell. The *legal integrator* (Appendix E) fits the same ridge target from the candidate’s own initial-frame embedding, recent and cumulative executed actions, and normalized time; it never reads a later frame, and it exists to expose how much of the endpoint a causally legal shortcut can already explain. Confidence intervals (CIs) use 100,000 crossed bootstrap draws resampling the task and seed axes independently, which preserves both generalization axes; an iid bootstrap over 25 cells would estimate a different quantity.

4.3 THE CONJUNCTIVE CONFIRMATION RULE

Confirmation was registered as a *conjunction* of eleven gates spanning integrity, the recurrent/no-carrier/integrator comparisons, deep-gap persistence, the component interventions of Table 1, a clean-prior guard, representation health (per-cell channel variance and effective rank), and late-training convergence. All thresholds, the metric, the aggregation, and the analysis code were frozen before any cohort result existed; any single miss fixes the immutable label `CONFIRMATION_FAILED`. The gates are enumerated with their frozen requirements and observed receipts in Table 2.

5 RESULTS: THE FROZEN CONJUNCTION FAILS

The write-once analysis validates **200/200** cells, all 100-epoch histories, rollouts, remote receipts, and source/cache/command hashes. Four gates pass and seven fail; the official label is **CONFIRMATION_FAILED**. Table 2 gives every gate receipt, and Figure 2 places every registered effect on one sign convention.

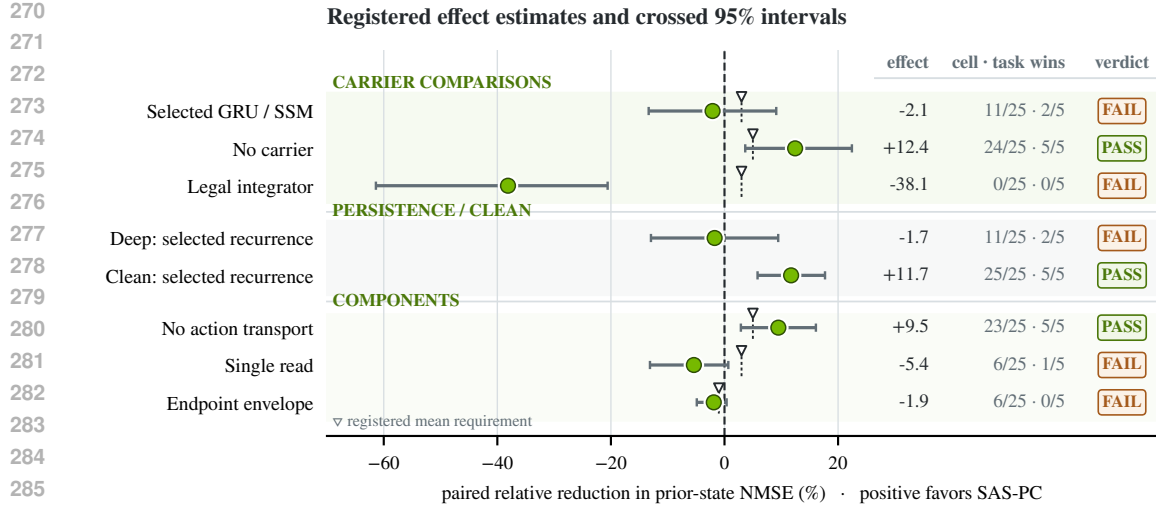


Figure 2: Registered effects on one sign convention. Green points are SAS-PC estimates, gray bars are crossed task-by-seed 95% intervals, and the right columns report estimates, cell/task wins, and gate verdicts. Positive favors SAS-PC; zero is an effect reference, not every row’s decision boundary. Table 2 gives the distinct frozen criteria; decisions use unrounded values.

Table 3: Held-out prior-state NMSE by task. Mean $\pm$ SD over five seeds; lower is better. Green identifies the candidate header; bold denotes the lowest mean across the four designs.

Task	Trained design			
	No carrier	GRU	Diag. SSM	SAS-PC
Acrobot	0.713 $\pm$ 0.027	0.685 $\pm$ 0.053	0.472 $\pm$ 0.019	0.572 $\pm$ 0.047
Manipulator	1.017 $\pm$ 0.025	0.978 $\pm$ 0.007	0.967 $\pm$ 0.016	0.959 $\pm$ 0.012
Quadruped	1.200 $\pm$ 0.195	1.064 $\pm$ 0.004	1.170 $\pm$ 0.177	1.097 $\pm$ 0.179
Stacker	0.961 $\pm$ 0.014	0.913 $\pm$ 0.008	0.895 $\pm$ 0.008	0.930 $\pm$ 0.020
Swimmer-15	1.222 $\pm$ 0.158	1.101 $\pm$ 0.042	1.079 $\pm$ 0.024	0.890 $\pm$ 0.045

Note. NMSE is standardized within task and is not pooled across tasks.

5.1 THE NO-CARRIER CONTRAST IS FAVORABLE; STRONGER REFERENCES ARE NOT

The no-carrier effect is broad (+12.44%, 24/25 cells, 5/5 tasks): within this grid, episode state descriptively beats the finite window. It is not a validated class or method-ranking result, because the stronger references reverse it. The per-cell recurrent envelope yields -2.10% (GRU selected 10 times, SSM 15), and the legal integrator is better in 25/25 cells and 5/5 tasks — pooled prior-state NMSE 0.889 for SAS-PC against 0.656 for the integrator. Even the integrator’s smallest task-level advantage corresponds to a SAS-PC effect of -15.91%, the largest -77.40%. The integrator is a diagnostic rather than a deployable world model, but the miss is neither marginal nor task-specific.

The recurrent comparison is structured rather than uniformly noisy (Figure 3a). Acrobot is -21.04% with 0/5 seed wins while Swimmer is +16.83% with 5/5; Quadruped’s task effect is only -3.02%, yet its cells range from -29.67% to +11.62%. The near-zero aggregate therefore combines stable task reversals with one genuinely seed-sensitive task; it is not evidence of task-invariant equivalence. Table 3 reports the underlying per-task errors.

Clean-state quality also fails to transfer to the corrupted endpoint. Against the same fixed recurrent identities, SAS-PC gains +11.73% (95% CI [+5.82%, +17.70%]; 25/25 cells) on the clean prior but -2.10% on the registered corrupted prior. Uniform clean decodability does not establish robust missing-observation transport.

The aggregate further hides a corruption reversal (Figure 3b). Freeze is +10.10% (25/25 cells, 5/5 tasks) and long freeze +10.10% (23/25, 5/5), whereas Gaussian noise is -20.95% (6/25, 1/5) and checkerboard -2.98% (9/25, 1/5). The complete carrier’s advantage is descriptively confined to replacement-style temporal freezes rather than generic corruption robustness — the pattern Equation 3 makes testable.

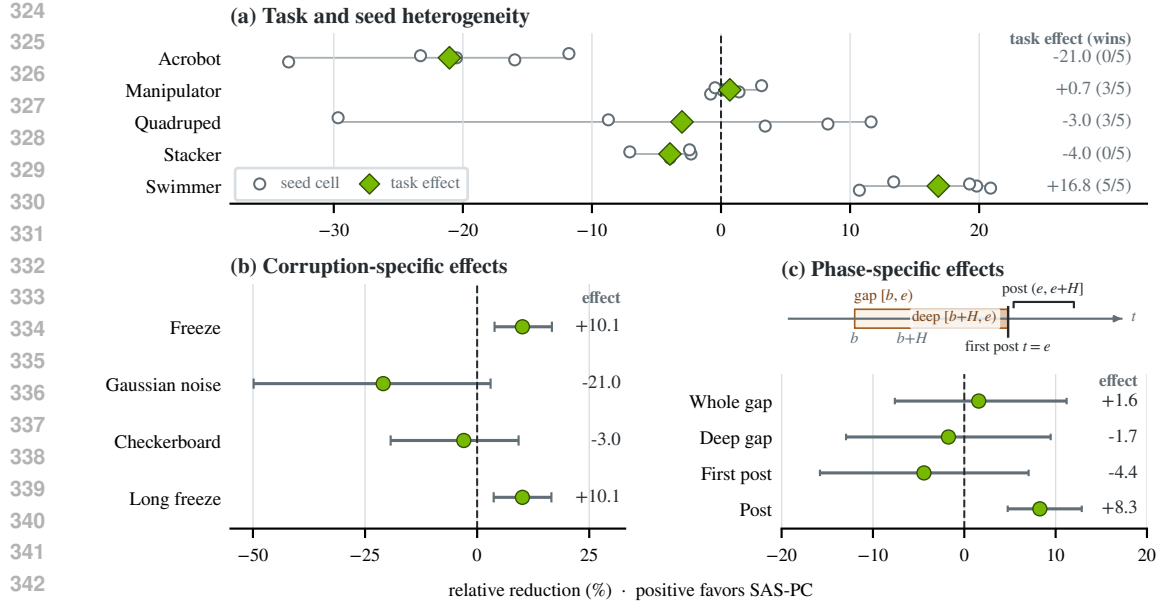


Figure 3: Where the aggregate changes sign. (a) Seed cells and task-mean effects against the primary-selected recurrent identity. (b) Corruption-specific and (c) equal-condition phase effects with descriptive crossed 95% intervals; the timeline glyph defines the phases. Positive favors SAS-PC. The post phase excludes the first reappearing observation; deep gap is nested within whole gap. Panels (b, c) are unadjusted, define no new gate, and cannot change ‘CONFIRMATION_FAILED’.

The phase profile sharpens the interpretation (Figure 3c). Deep-gap (-1.74%, 11/25) and first-post (-4.42%, 10/25) effects are unfavorable, but the later post-gap effect is +8.27% (95% CI [+4.75%, +12.86%]; 23/25 cells, 5/5 tasks). Descriptively, the candidate looks most useful during recovery after observations resume — not at the deepest missing-observation or first-reappearance points where a persistent-transport story would predict an advantage.

5.2 ACTION TRANSPORT PASSES ITS CONTRAST; HIERARCHY CONTRASTS DO NOT

Removing recurrent action features (the no-action arm) worsens all five task means and 23/25 cell effects, giving +9.48% (95% CI [+2.88%, +16.09%]) and passing its registered contrast; its descriptive effect stays positive under every corruption, from +5.30% to +10.48%. The complete carrier’s recurrent-baseline reversal is therefore narrower than the contribution of action transport itself: the mechanism of Equation 2 meets its registered bar within this grid even where the full design does not.

The hierarchy evidence points the other way. Joint access to both states yields -5.39% (95% CI [-13.12%, +0.67%]) with only 6/25 wins; it is near neutral on freeze (+0.50%) and long freeze (+1.65%) but unfavorable on Gaussian (-12.57%) and checkerboard (-8.00%). Learned shrinkage loses every task mean to the per-cell better fixed- ρ endpoint (-1.89%; dynamic selected 18 of 25 cells, static 7), so the interior point of Equation 3 was not vindicated either. Consistently, the simpler single-read control has the best within-block rank profile of all eight designs — mean rank 2.20 (first in 10/25, top-three in 20/25) versus 3.80 for full SAS-PC (3/25; 10/25; Figure 4, Appendix C). These ranks are descriptive, but they reinforce rather than rescue the failed component contrasts.

The optimizer-seed axis makes the same separation. Equal-task effects are positive at all five seeds for no carrier and action transport, but at only 2/5 seeds for the recurrent envelope and 0/5 for joint read. This consistency check is not a gate; it shows the favorable class and action patterns are less seed-dependent than the SAS-PC-specific and hierarchy comparisons.

5.3 VALIDITY FAILURES CONCENTRATE BY TASK AND SPAN ARMS

Variance passes 200/200 cells, but effective rank passes only 144/200 and convergence 126/200. The strongest favorable recurrent task, Swimmer (+16.83%, 5/5 wins), has 0/40 joint guard passes; the

strongest unfavorable task, Acrobot (-21.04%, 0/5), has 40/40 (Table 4). Yet joint-pass counts are nearly identical across designs — 14/25 to 15/25, with SAS-PC at 15/25 — so the health problem is task-structured rather than specific to one arm (Figure 4). We do not filter to the 118 jointly passing cells: that would redefine the frozen estimand after seeing outcomes.

Table 4: Representation-rank and convergence receipts by task. Counts aggregate eight designs $\times$ five seeds.

Task	Rank ≥ 16	Change $\leq 5\%$	Both guards
Acrobot	40/40	40/40	40/40
Manipulator	40/40	39/40	39/40
Quadruped	24/40	5/40	4/40
Stacker	40/40	35/40	35/40
Swimmer-15	0/40	7/40	0/40
Overall	144/200	126/200	118/200

These diagnostics leave an artifact-complete falsification but block promotion of the within-grid rankings to claims about a generally superior healthy, converged system. Appendix B reports all registered contrasts in full; Appendix C gives all eight raw designs, exact ranks, and condition/phase receipts; Appendix D gives arm-level health and convergence counts.

6 DISCUSSION AND LIMITATIONS

A class benefit is not a carrier benefit. A causal $H = 3$ predictor has no episode state, so the favorable no-carrier contrast is compatible with older evidence being useful. The stronger recurrent envelope asks a different question: does *this* shared-action predict-correct carrier use that opportunity better than ordinary learned recurrence? Its negative estimate prevents that promotion. The distinction matters because a finite-window baseline can make any recurrent module look like evidence for its own internal design; only the registered comparators can establish that SAS-PC is the right implementation.

The integrator changes how the endpoint should be read. Under random-action control trajectories, an initial visual embedding plus executed actions and time retains substantial information about native state. Losing all 25 integrator comparisons does not mean later images are useless; it means the chosen prior-state endpoint admits a strong causally legal shortcut, and SAS-PC does not extract more decodable state than that summary. Future memory benchmarks should report both a recurrent visual baseline and an initial-frame/action integrator; beating only the finite-window host leaves the central ambiguity unresolved.

Predict-correct behavior is corruption-dependent. Freezes preserve the last visual value while removing new evidence; Gaussian and checkerboard corrupt the current input itself. The sign reversal is consistent with the hypothesis embedded in Equation 3: action prediction helps when observations are absent, while correction can hurt when the incoming latent is unreliable and the learned gate fails to close. The phase profile sharpens this — the favorable descriptive effect appears after observations resume, resembling improved recovery more than superior blind propagation. It remains a hypothesis because condition and phase intervals are unadjusted and the exporter did not retain per-step gate trajectories.

Clean decodability is not persistent transport. SAS-PC improves every clean-prior cell against the fixed recurrent identities while losing the corresponding corrupted-prior contrast. A clean objective can strengthen latent statistics or ordinary one-step prediction without preserving the right information through missing observations. Memory studies should therefore report clean prior, blind-gap prior, first-reappearance correction, and later recovery separately; a single clean linear-probe gain cannot stand in for memory robustness.

The two-state hierarchy is not supported. Action transport is the only component intervention meeting its registered bar, and its effect spans all four corruptions. In contrast, the single-read control has the best rank profile, the joint-read contrast is unfavorable, and learned interior shrinkage loses to the endpoint envelope. The data support neither a necessary fast/medium decomposition nor learned horizon discovery: here the extra routed read behaves as complexity without demonstrated benefit, though multiple state variables may still help in other regimes.

Validity is part of the estimand, not a cleanup step. Swimmer supplies the strongest favorable recurrent effect while failing the rank guard in every arm; Acrobot supplies the strongest unfavorable effect while passing everywhere. Because joint-pass counts are nearly constant across designs, the pathology is not a convenient excuse for one candidate’s loss — it is a task-level warning about what the whole grid measures. Filtering to healthy cells would answer a different question, so the negative label retains all 200 cells.

Design implication for a new cohort. The next study should combine an exact-SIGReg LeWM host with representation stabilization, an action-conditioned *single-state* predict–correct baseline, internally action-conditioned GRU/SSM comparators, and the same legal integrator; it should retain per-step correction gates and route weights, preregister deep-gap versus recovery-phase predictions, and evaluate executed control or planning in addition to state probes. Those changes test the narrow hypotheses exposed here — action transport and recovery — instead of rerunning the rejected broad claim.

Limitations remain substantial. The host uses VICReg-style clean-target regularization rather than original SIGReg LeWM; corruptions and random actions emphasize state estimation, not return. GRU matching is approximate, and the comparison does not exhaust RSSM, S4/Mamba, retrieval, or long-context baselines. SAS-PC was selected adaptively on other tasks. The exporter retained only final shrinkage coefficients and action-feature norms; **per-step gate vectors and route weights were not retained**, so no trajectory-level claim is possible. Any replay is post hoc and cannot alter the frozen result; all proposed extensions require a genuinely new cohort.

7 CONCLUSION

Finite context and persistent state are different architectural properties, but a favorable no-carrier contrast is insufficient to validate a recurrent design. Across 200 frozen cells, the no-carrier and action-transport contrasts favor SAS-PC; the stronger recurrent and integrator references do not; the hierarchy contrasts are unsupported; and the validity guards fail. The complete negative result is more informative than a baseline win: it falsifies the broad method claim on this cohort while isolating action transport — the one mechanism that survived its own intervention — as the hypothesis a new, valid cohort should test.

REPRODUCIBILITY STATEMENT

The anonymous supplement contains the frozen protocol and source, all 200 cell rows, 33 registered contrasts, task/seed effect matrices, figures, and deterministic verification tools. Hash-bound private artifacts additionally retain checkpoints, histories, held-out rollouts, and remote receipts. Two process-level interruptions were audited before final analysis. The first occurred with 136 valid cells and required four complete-cell restarts; the second occurred with 180 valid cells and required one. The schema-v2 audit verifies interrupted and replacement logs, local and remote W&B terminal states, replacement artifact hashes, terminal runner quiescence, and the final attempts, runs, summary, CSV, and analysis bytes; all five replacements reached epoch 100. Appendix F itemizes the audited interruptions. Full identities are redacted for review; the curated archive can be verified without the private repository.

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A EXACT PROTOCOL AND ANALYSIS

A.1 OBJECTIVE TERMS, INITIALIZATION, AND PARAMETER MATCHING

The variance and covariance terms of Equation 1 act on the matrix $\bar{Z}^*$ of batch-and-time-centered active clean targets with N rows:

$$\mathcal{L}_{\text{var}}(Z^*) = \frac{1}{D} \sum_{d=1}^D \max(0, 1 - \sqrt{\text{Var}(z_{\cdot d}^*) + \varepsilon}), \quad \mathcal{L}_{\text{cov}}(Z^*) = \frac{1}{D} \sum_{d \neq d'} C_{dd'}^2, \quad C = \frac{\bar{Z}^{*\top} \bar{Z}^*}{N-1}. \quad (6)$$

All three loss terms have unit weight; there is no selectable regularizer coefficient. Clean-frame activations, simulator state, rewards, and corruption labels or masks never enter the recurrent update; there is no memory teacher and no memory-specific loss. SAS-PC initialization is $W_x = I$, $W_o = W_a = 0$, $b_f = b_m = 2$, $c_f = c_m = 0$, uniform route. The GRU width is fixed once by minimizing $|4Dh + 3h^2 + 6h - (2D^2 + 16D)|$ over hidden sizes $h \leq D$, never re-selected per task.

A.2 IMMUTABLE IDENTITIES

The frozen execution identities are:

Table 5: Frozen execution and artifact identities.

Identity	Frozen value or SHA-256 prefix
clean worktree at freeze	True
protocol SHA-256	357cbe1296926802...
canonical public command-list SHA-256	408051d027c55ae5...
cache-manifest SHA-256	81e3efc64b822741...
cache-sidecar SHA-256	dbc2db0700c30a3e...
artifact-manifest SHA-256	1e3bbcee667a98a6...
cell CSV SHA-256	71144b471acbf88a...
contrast CSV SHA-256	78b804c3acfc433c...

The four-GPU queue was fixed as GPU 0: Acrobot then Stacker; GPU 1: Manipulator; GPU 2: Quadrupe; GPU 3: Swimmer. Every task used optimizer seeds 18001–18005. The protocol stores all 200 expanded commands and per-source SHA-256 values; the final analyzer revalidates them before writing results.

A.3 METRIC PHASES

For target time t and corruption interval $[b, e)$, gap is $b \leq t < e$, deep is $b + H \leq t < e$, first_post is $t = e$, and post is $e < t \leq e + H$. The primary held-out metric selects deep and first-post samples. This selection, the four-corruption average, and all probe splits were frozen before launch.

A.4 CROSSED BOOTSTRAP

Let $\Delta \in \mathbb{R}^{5 \times 5}$ contain task-by-seed paired effects. Each draw samples five task indices and five seed indices independently with replacement, takes their Cartesian 5×5 submatrix, and averages all entries. We use 100,000 PCG64 draws with seed 18018 and linear 2.5/97.5 percentiles, preserving the two crossed generalization axes.

A.5 COMPARATOR IDENTITY

For each task–seed cell, the recurrent identity is the lower primary held-out prior NMSE of GRU and SSM; exact ties select GRU. The same identity supplies deep and clean references; selecting a new best model separately for those metrics is prohibited. Static/dynamic identity is selected per cell only for the endpoint noninferiority contrast.

A.6 FROZEN ARMS AND PARAMETER RECEIPTS

The eight arms of Table 1 are separately trained; GRU/SSM update from visual latents only, although the shared predictor remains action-conditioned. SAS-PC has 33,286–36,614 carrier parameters across action dimensions; the fixed GRU width gives 35,048. Recurrent carrier sizes differ by at most 5.29% and total models by at most 0.09%.

B REGISTERED CONTRASTS IN FULL

All eight registered primary contrasts with crossed 95% and 90% intervals and pooled absolute errors:

Table 6: Registered held-out primary contrasts with 95% and 90% crossed bootstrap intervals and pooled candidate/reference NMSE. Deep gap and clean prior use the deep and clean prior-state NMSE against the recurrent envelope; action transport and joint read compare against the no-action and single-read controls; the endpoint envelope is the shrinkage gate of Table 2.

Contrast	Effect	95% CI	90% CI	cells	tasks	cand./ref. NMSE
Recurrent envelope	-2.10%	[-13.35%, +9.11%]	[-11.55%, +7.36%]	11/25	2/5	0.889/0.893
No carrier	+12.44%	[+3.65%, +22.43%]	[+4.89%, +20.70%]	24/25	5/5	0.889/1.023
Legal integrator	-38.12%	[-61.38%, -20.58%]	[-56.84%, -21.91%]	0/25	0/5	0.889/0.656
Deep gap	-1.74%	[-12.95%, +9.46%]	[-11.15%, +7.70%]	11/25	2/5	0.883/0.891
Clean prior	+11.73%	[+5.82%, +17.70%]	[+6.81%, +16.92%]	25/25	5/5	0.738/0.843
Action transport	+9.48%	[+2.88%, +16.09%]	[+3.94%, +15.02%]	23/25	5/5	0.889/1.000
Joint read	-5.39%	[-13.12%, +0.67%]	[-11.64%, +0.07%]	6/25	1/5	0.889/0.855
Endpoint envelope	-1.89%	[-4.88%, +0.37%]	[-4.22%, +0.00%]	6/25	0/5	0.889/0.876

The table is generated directly from `confirmation_analysis.json`; decisions use full precision, so rounding here cannot change a gate. All 200 auditable cell rows, the 33 registered contrast rows, the 5×5 cell-effect matrices, and selected identities are supplied in `confirmation_cells.csv`, `confirmation_contrasts.csv`, and `confirmation_analysis.json`.

C FULL RESULTS

Table 3 and Table 7 jointly report all eight trained designs. Values are held-out prior-state NMSE as mean $\pm$ standard deviation over five optimizer seeds; lower is better.

Table 7: Held-out prior-state NMSE for SAS-PC component and endpoint controls by task (mean $\pm$ SD over five seeds; lower is better).

Task	No action	Single read	Static	Dynamic
Acrobot	0.619 $\pm$ 0.174	0.483 $\pm$ 0.054	0.609 $\pm$ 0.118	0.548 $\pm$ 0.037
Manipulator	1.026 $\pm$ 0.030	0.953 $\pm$ 0.013	0.957 $\pm$ 0.011	0.958 $\pm$ 0.011
Quadruped	1.304 $\pm$ 0.214	1.019 $\pm$ 0.137	1.131 $\pm$ 0.188	1.120 $\pm$ 0.231
Stacker	0.958 $\pm$ 0.027	0.905 $\pm$ 0.013	0.932 $\pm$ 0.026	0.921 $\pm$ 0.020
Swimmer-15	1.092 $\pm$ 0.050	0.916 $\pm$ 0.038	0.991 $\pm$ 0.142	0.878 $\pm$ 0.010

Table 8: Clean prior-state NMSE by task (mean $\pm$ SD over five seeds; lower is better). Bold denotes the lowest mean per task.

Task	No carrier	GRU	Diag. SSM	SAS-PC
Acrobot	0.410 $\pm$ 0.016	0.462 $\pm$ 0.014	0.364 $\pm$ 0.020	0.329 $\pm$ 0.014
Manipulator	0.921 $\pm$ 0.005	0.925 $\pm$ 0.002	0.927 $\pm$ 0.005	0.879 $\pm$ 0.006
Quadruped	1.045 $\pm$ 0.009	1.054 $\pm$ 0.002	1.057 $\pm$ 0.003	0.841 $\pm$ 0.029
Stacker	0.868 $\pm$ 0.006	0.863 $\pm$ 0.007	0.862 $\pm$ 0.006	0.821 $\pm$ 0.003
Swimmer-15	0.980 $\pm$ 0.002	1.008 $\pm$ 0.001	1.016 $\pm$ 0.001	0.818 $\pm$ 0.012

Exact rank summaries are:

Table 9: Descriptive within-task, within-seed rank summaries corresponding to Figure 4.

Design	Mean rank	Median	First	Top-3
Single read	2.20	2	10/25	20/25
Dynamic	3.12	3	3/25	18/25
Diag. SSM	3.72	5	7/25	12/25
SAS-PC	3.80	4	3/25	10/25
Static	4.24	4	1/25	9/25
GRU	4.96	5	1/25	5/25
No action	6.84	7	0/25	0/25
No carrier	7.12	7	0/25	1/25

The frozen recurrent-envelope identity counts by task are:

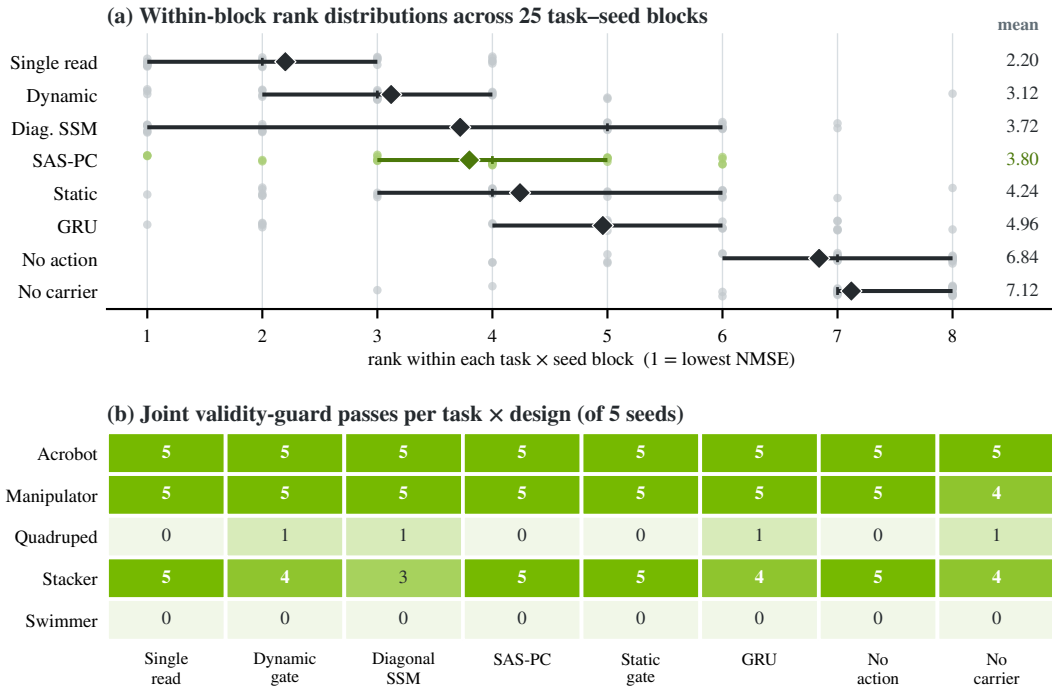


Figure 4: (a) Within-task, within-seed rank distributions for all eight designs; light points are the 25 task-seed ranks, bars span the interquartile range, ticks mark medians, diamonds mark mean rank, and rank 1 is the lowest NMSE within a block. SAS-PC is highlighted only to locate the candidate; ranks are descriptive, pool no raw NMSE across tasks, and define no superiority test. (b) Joint validity-guard passes per task and design (of five seeds), showing that guard failures are task-structured rather than design-specific.

Table 10: Recurrent-envelope identities selected on the primary metric and reused for deep, clean, task, and corruption slices.

Task	GRU selected	SSM selected
Acrobot	0/5	5/5
Manipulator	2/5	3/5
Quadruped	5/5	0/5
Stacker	0/5	5/5
Swimmer-15	3/5	2/5

Condition-specific component effects and equal-condition phase slices provide secondary context for Figure 3:

Table 11: Descriptive corruption-specific component effects. Positive values favor full SAS-PC; no multiplicity correction or decision gate is defined.

Corruption	vs no carrier	action transport	joint read
Freeze	+10.91%	+9.79%	+0.50%
Gaussian noise	+16.68%	+10.48%	-12.57%
Checkerboard	+5.70%	+5.30%	-8.00%
Long freeze	+10.11%	+9.44%	+1.65%

Table 12: Descriptive phase effects against the primary-selected GRU/SSM identity, averaged equally across corruptions. Deep gap is nested within whole gap; intervals are unadjusted and define no decision gate.

Phase	Effect [95% CI]	Cell/task wins
Whole gap	+1.58% [-7.60%, +11.20%]	13/25; 3/5
Deep gap	-1.74% [-12.95%, +9.46%]	11/25; 2/5
First post-gap	-4.42% [-15.80%, +7.04%]	10/25; 1/5
Post-gap	+8.27% [+4.75%, +12.86%]	23/25; 5/5

Per-corruption and per-phase values remain in each cell’s `metrics.json` and held-out rollout arrays. The paper does not pool raw task-state MSE across heterogeneous tasks.

D REPRESENTATION AND CONVERGENCE

Table 13: Representation-health receipts by design.

Design	Min. variance	Min. rank	Variance pass	Rank pass
No carrier	0.0296	2.92	25/25	18/25
GRU	0.0230	2.02	25/25	17/25
Diag. SSM	0.0309	3.81	25/25	19/25
SAS-PC	0.0228	2.02	25/25	18/25
No action	0.0233	2.03	25/25	18/25
Single read	0.0245	2.38	25/25	18/25
Static	0.0349	4.62	25/25	18/25
Dynamic	0.0233	2.02	25/25	18/25

Table 14: Late-window convergence receipts by design.

Design	Largest late change	Convergence pass
No carrier	+39.38%	15/25
GRU	+43.71%	15/25
Diag. SSM	+47.70%	16/25
SAS-PC	+90.25%	15/25
No action	+132.09%	17/25
Single read	+120.28%	15/25
Static	+36.58%	17/25
Dynamic	+47.34%	16/25

Representation health requires both thresholds to be met in every cell. Convergence is the absolute relative change between mean validation predictive loss over epochs 81–90 and 91–100. The gate requires every cell’s late change to be at most 5%; no failing arm is removed.

E INTEGRATOR GUARD

For target t , the legal feature vector is $[E_\theta(o_0), a_{t-3:t-1}, \sum_{j=0}^{t-1} a_j, t/(L-1)]$, with zero-padding before three actions exist. The candidate checkpoint supplies E_θ ; a ridge map is fit on clean training targets and evaluated only on deep and first-post held-out samples. The integrator receives no observation after the visible o_0 .

Table 15: Legal initial-frame/action integrator guard by task. Positive paired reduction favors SAS-PC.

Task	SAS-PC prior NMSE	Integrator NMSE	Paired reduction	Seed wins
Acrobot	0.572	0.416	-37.45%	0/5
Manipulator	0.959	0.821	-16.74%	0/5
Quadruped	1.097	0.618	-77.40%	0/5
Stacker	0.930	0.802	-15.91%	0/5
Swimmer-15	0.890	0.622	-43.14%	0/5

F ADAPTIVE PROVENANCE AND RESTART

SAS-PC was selected after V1–V7 development on different tasks. Its earlier adaptive V8 study completed 325 cells but had immutable label `PILOT_NO_GO_FINAL_DESCRIPTIVE`; it cannot rescue the present confirmation (V18). V9–V17 are likewise development/host audits, not confirmation. Interruption 1, `acrobot.swingup/vicreg_ssm/s18005`: last interrupted epoch 62; interrupted/replacement log hashes `dfe369fac5d9/de69a7091d8a`; replacement epochs 1–100. Interruption 1, `manipulator.bring_ball/vicreg_ssm/s18005`: last interrupted epoch 31; interrupted/replacement log hashes `3492682adb9/2c516e0b6f8c`; replacement

epochs 1–100. Interruption 1, `quadruped.run/vicreg_ssm/s18005`: last interrupted epoch 9; interrupted/replacement log hashes `6a72339d3e8e/6f005ea3ae51`; replacement epochs 1–100. Interruption 1, `swimmer.swimmer15/vicreg_ssm/s18005`: last interrupted epoch 62; interrupted/replacement log hashes `4029c6a1b1be/5e0815242313`; replacement epochs 1–100. Interruption 2, `stacker.stack_4/vicreg_hacssmv8_static/s18003`: last interrupted epoch 37; interrupted/replacement log hashes `e74f6ba0dcb1/63b9ee95f0eb`; replacement epochs 1–100. The runner’s JSON attempts ledger records terminal subprocess returns, so process-killed attempts are supplied by this independently verified schema-v2 lineage receipt. The audit’s separately classified parent-only pause did not signal the trainer and is not counted as an additional restart. The registered commands, seeds, source hashes, and stopping rule were unchanged.

G CLAIM BOUNDARY

The study can establish only a persistent-state or component-intervention result for a VICReg-trained LeWM-derived finite-context host on the frozen corruption cohort. It cannot establish improvement to original SIGReg LeWM, executed-return or planning gains, causal discovery, learned semantic hierarchy, calibrated uncertainty, learned horizon discovery, or robustness beyond these tasks and corruptions.

H LLM USAGE STATEMENT

OpenAI Codex assisted with code review, experiment monitoring, artifact auditing, deterministic result-to-manuscript tooling, and manuscript drafting/editing. The authors verified the executed code, artifacts, statistics, citations, and final claims and retain responsibility for the work.